Causal Diagrams & Model Planning

June 2025

Old System:

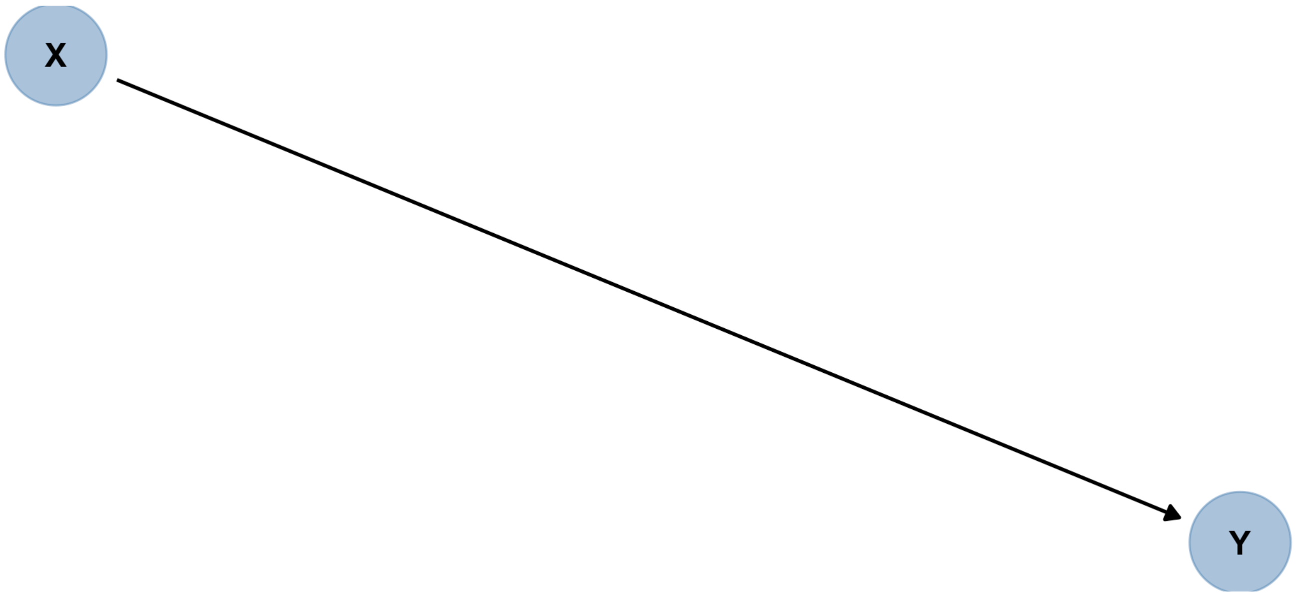
"Include every predictor you think is relevant and important"

(Data) Size Matters!

- $p < \frac{n}{15}$: *Roughly* 10-20 "events" per parameter estimated (Harrell 2015)
- Assumes even dist. over categories
- Consider: domain expertise, data availability, missing-ness, dynamic range...& *expected causal relationships*

Causal Diagrams

There's more to planning than just p < n/15!



"Causal"?

Can we actually make *causal conclusions*?

- If detect association, can conclude ___ *causes* ___ IFF...

- Otherwise:

- So why use "causal" diagrams to plan models?
  - Transparency about assumptions
  - Clear, consistent way to choose predictors

H.E.A.L.T.H. Camp Example

Health Education And Leadership Training for a Hopeful future (H.E.A.L.T.H) Camp "is a free weeklong summer camp for young people from mostly medically underserved communities in Grand Rapids to learn more about their own health and health careers at Calvin U."

H.E.A.L.T.H. Camp Example

Response: Test Score

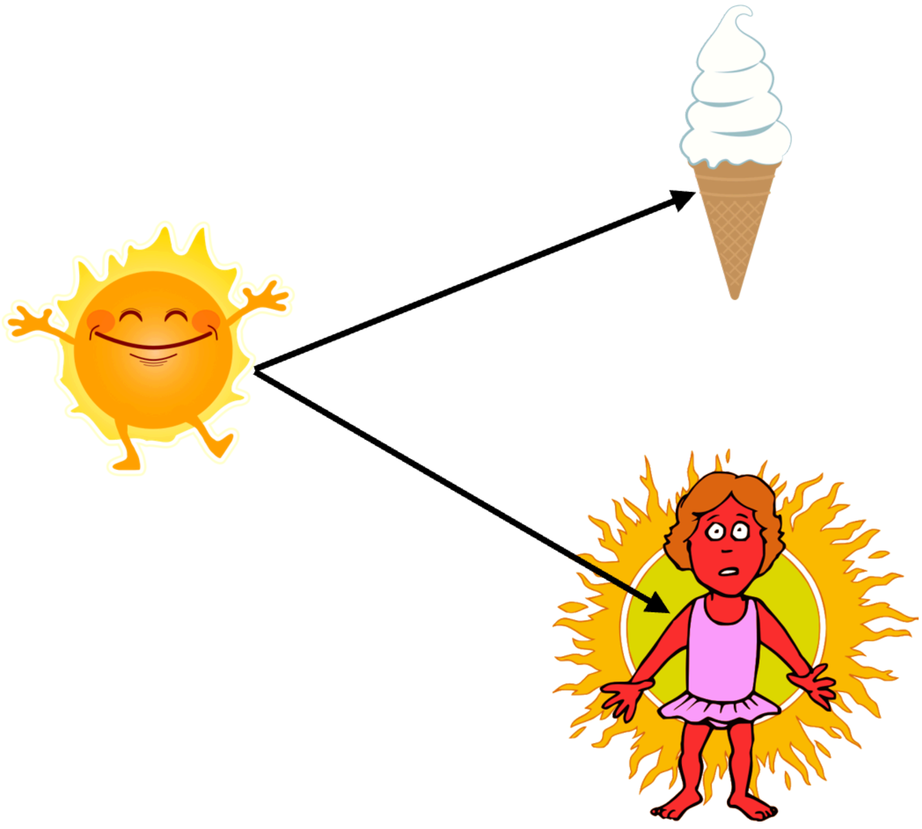
- Age
- Sex
- Grade in school
- Neighborhood
- Race & Ethnicity
- Income
- Interest in health career
- Timepoint (before/after camp)

Classify to Plan

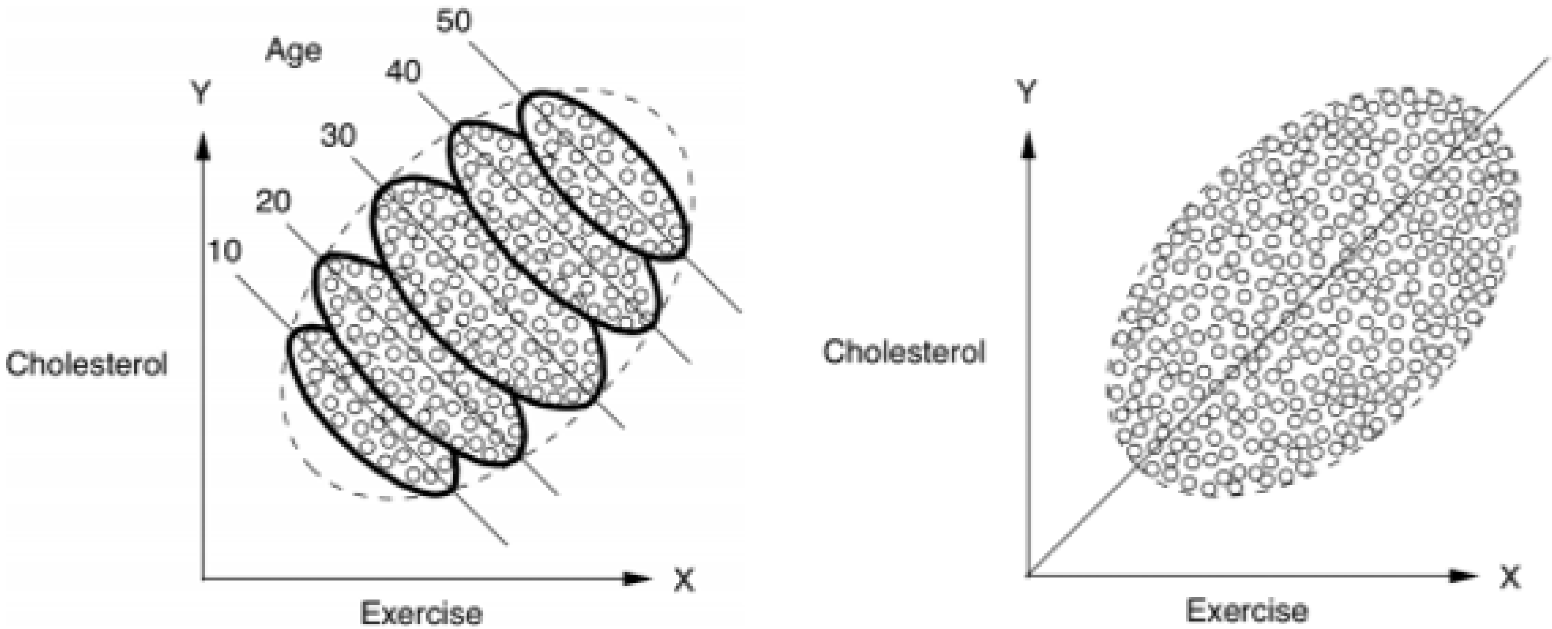
- *First* choose a key predictor whose effect you'll quantify
- Determine other candidate predictors' relative roles
- Include/exclude in model depending on this classification

Confounder

a.k.a. "common cause"



Confounder



But...What's *wrong* with this picture?

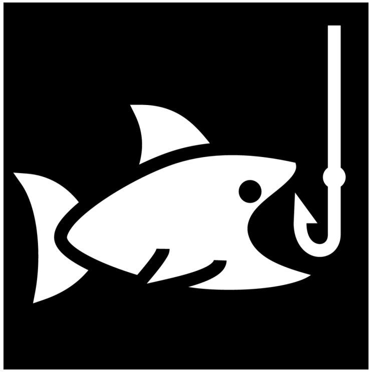
Image: https://www.causa.tech

Precision Covariate

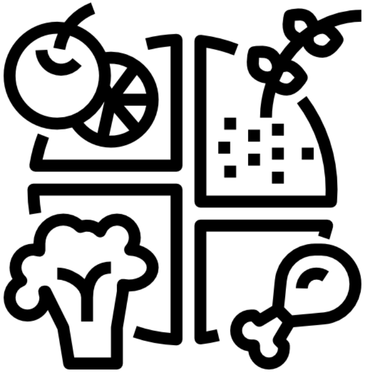
Mediator

Collider

U. MN St Paul Campus



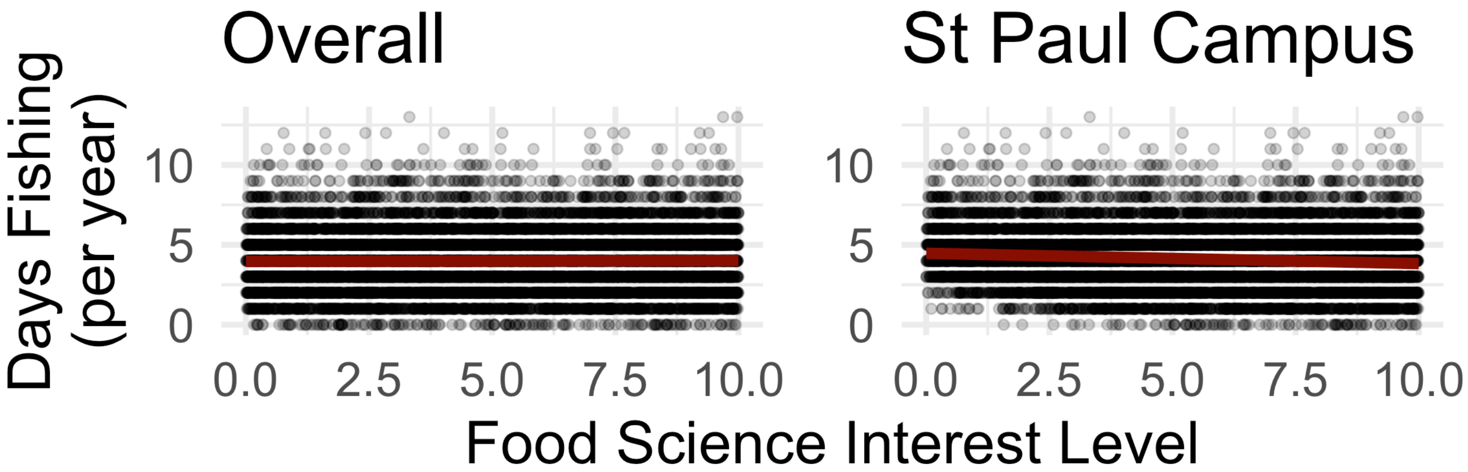
Food Science Interest



Days Fishing

Collider

Fishing *or* Food -> St Paul



Example from [J. Fieberg]

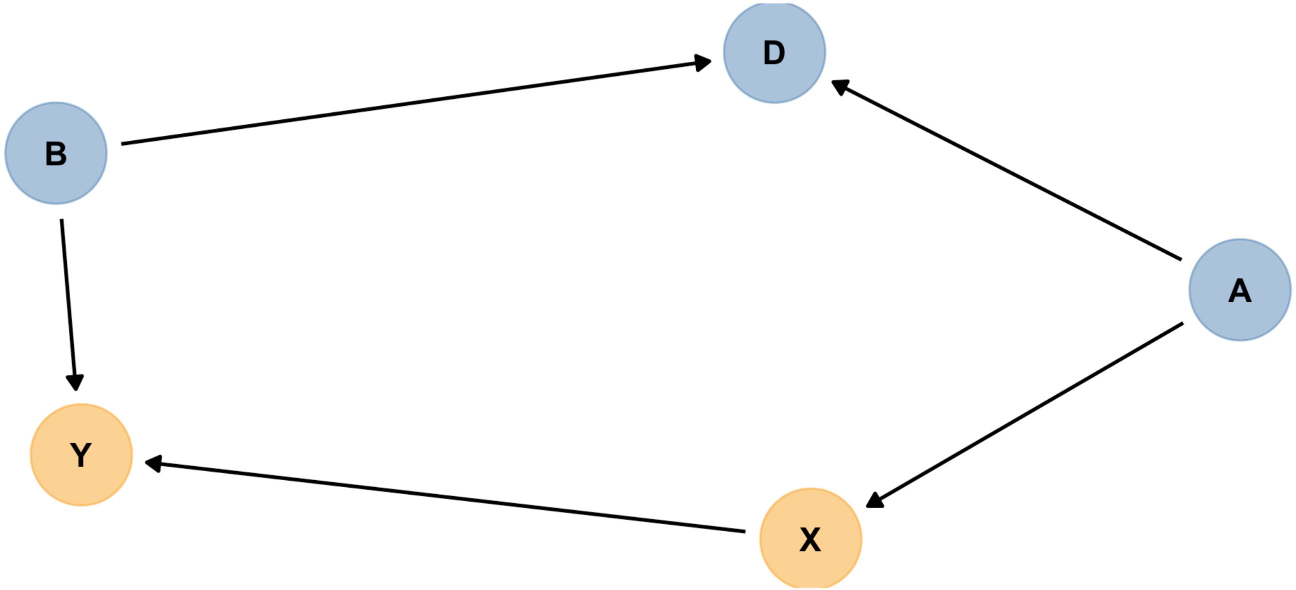
Collider

Moderator or Modifier

Also known as: *Interaction*

Can get complex

M-Bias



We've chosen some simple principles but there could be much more to learn!

Resource: Guide to Causal Inference https://doi.org/10.1098/rspb.2020.2815

Look back

which variables should be predictors to model HEALTH Camp test score?

score ~

Maybe see: [Model Planning Checklist]

Going Further

References & Resources

- A biologist's guide to model selection and causal inference (Laubach et al.)
- *Statistics for Ecologists* (J. Fieberg) Ch. 7
- *Statistical Modeling* (D. Kaplan) Ch. 17
- An Introduction to Causal Inference (F. Dablander)
- and maybe, *The Book of Why* (J. Pearl & D. Mackenzie, ISBN: 978-0465097609)

What's Your Model Process?

Feedback

https://cutt.ly/arm-feedback